Fiscal Year:	FY 2004	**Task Last Updated:**	FY 09/29/2004
PI Name:	Peterson, Kevin Ph.D.		
Project Title:	Using Molecular Clocks And Regulatory Networks To Test Origin-Of-Bilateria Scenarios		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Evolutionary biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Musculoskeletal Biology		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-06
Start Date:	11/01/2002	**End Date:**	10/31/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG2-1584		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Because of the wealth of both phylogenetic and developmental data, there is now a tremendous interest in the origin of bilaterian metazoans and the associated paleontological events such as the Cambrian explosion. It is now generally accepted that the latest common ancestor of bilaterians was a fairly complex animal complete with anterior-posterior and dorsal-ventral axes, and a central nervous system. Nonetheless, it is still unclear if the origin of this latest common ancestor shortly preceded the Cambrian explosion, as suggested by the fossil record, or if there was a significant but cryptic period of bilaterian evolution before their first appearance in the fossil record, as suggested by most molecular-clock studies. Understanding which stage of the bilaterian life cycle is primitive for Bilateria, the larval stage or the adult stage, is crucial for deciphering the temporal incongruence between the fossil record and these molecular-clock analyses. To test for homology of larval types, we cloned a gene in the purple sea urchin		

Task Description:	Strongylocentrotus purpuratus called NK2.1, a member of the NK-2 class of homeodomain transcription factors. NK2.1 is expressed at the apical pole of the sea urchin, and transcripts are eventually restricted to the larval apical organ. Because the apical organ is restricted to primary larval forms, homology of the apical organ implies homology of the larva. In order to test this homology, we want to test for the conservation of gene batteries underlying apical pole development in sea urchins and in gastropod molluscs. We have already identified several important upstream factors in the sea urchin, and are in the process of examining these same genes in the mollusc. Determining the polarity of the larval vs. adult bilaterian body plan will then allow for a better understanding of major events in the history of life on Earth including the Cambrian explosion.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Ascertaining the primitive life cycle of bilaterians will shed tremendous insight into the ecological ramifications of surviving catastrophic environmental perturbations such as "snowball earth". Indeed, current discussions among snowball-earth workers have focused, in part, on the inferred ecology of the animals surviving the glaciation(s) (see Runnegar 2000, and discussion by Schrag & Hoffman 2001 and Hyde et al. 2001). Thus, a proper understanding of the earliest stages of bilaterian evolution is necessary in order to properly understand the geological and paleontological events of the Late Neoproterozoic. Finally, given that bilaterians may have been present for ~ 100 Ma before they made their first appearance in the fossil record, small but potentially complex metazoans may have thrived and diversified without leaving a trace on our own planet in a relatively recent time frame. Discussions centered on discovering complex life throughout the universe should keep this most interesting possibility in mind.
Task Progress:	"Using Gene Regulatory Networks to Test Origin-of-Bilateria Scenarios" NAG2-1584 P.I. Kevin J. Peterson Dartmouth College 10/6/03-9/20/04 Subject Inventions – NONE This year we have continued to focus our attention on "Hypothesis 1: Expression and Regulation of NK2.1 at the Apical Pole is Conserved in Polychaetes and Deuterostomes." We have completed the following: 1) The characterization and general upstream regulation of SpNK2.1 was published this year in Developmental Biology (Takacs et al. 2004. Expression of an NK2 homeodomain gene in the apical ectoderm defines a new territory in the early sea urchin embryo. Developmental Biology 269:152-164). 2) We have continued our work on cloning out the relevant genes from the gastropod mollusc Haliotis (the red abalone). We have isolated fragments of all of the genes of interest, constructed a lambda cDNA library, and are in the process of isolating the full-length cDNA sequences. We hope to begin in situ hybridizations soon. 3) We have isolated a potential downstream target of SpNK2.1, an isoform of alpha-tubulin which we have shown is exclusively expressed in the apical domain, and injection of the SpNK2.1 MASO abrogates its expression. We are in the process of isolating the promoter region and cloning the constructs for injection this winter. We have also isolated a fragment of the alpha-tubulin gene from Haliotis, and will search for isoforms in the lambda library soon. 4) We have also isolated six novel genes using subtractive-hybridization screens designed specifically to look for downstream targets of SpNK2.1. We are now determining expression profiles of these genes, and, if the expression patterns are apical, will begin the MASO injections this winter. Any putative downstream target will then be cloned from Haliotis. Scholarly Activities: Invited Speaker: November 2003: GSA Pardee symposium on Neoproterozoic Geobiology. Geological Society of America Annual Meeting, Seattle, WA. Organized by Shuhai Xiao and Jay Kaufman. April 2004: CSEOL Symposium The Origin of Animals. University of California, Los Angeles. January 2004: Department of Earth Sciences, Dartmouth College. September 2004: Department of Biology, University of Central Florida, Orlando.
Bibliography Type:	Description: (Last Updated: 02/23/2007)
Articles in Peer-reviewed Journals	Takacs, C.M.; Amore, G.; Oliveri, P.; Poustka, A.J.; Wang, D.; Burke, R.D.; Peterson, K.J. "Expression of an NK2 homeodomain gene in the apical ectoderm defines a new territory in the early sea urchin embryo." Developmental Biology 269(1):152-164, May 2004. , May-2004
Articles in Peer-reviewed Journals	Peterson, K.J. "Isolation of Hox and Parahox genes in the hemichordate Ptychodera flava and the evolution of deuterostome Hox genes." Molecular Phylogenetics and Evolution 31(3):1208-1215, Jun 2004. , Jun-2004
Articles in Peer-reviewed Journals	Peterson, K.J.; Lyons, J.B.; Nowak, K.S.; Takacs, C.M.; Wargo, M.J.; McPeek, M.A. "Estimating metazoan divergence times with a molecular clock." Proceedings of the National Academy of Sciences USA 101(17):6536-6541, Apr 2004. , Apr-2004
Books/Book Chapters	Smith, A.B.; Peterson, K.J.; Wray, G.; Littlewood, D.T.J. "From bilateral symmetry to pentaradiality: The phylogeny of hemichordates and echinoderms." in "Assembling The Tree of Life." Ed. J. Cracraft and M.J. Donoghue. Oxford University Press, New York, Jun 2004, p. 365-383. , Jun-2004
Books/Book Chapters	Eernisse, D.J.; Peterson, K.J. "The History of Animals." in "Assembling The Tree of Life." Ed. J. Cracraft and M.J. Donoghue. Oxford University Press, New York, Jun 2004, p. 197-208. , Jun-2004